Linear Statistical Models

B. Statistical Data Science 2nd Year Indian Statistical Institute

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Exercise Series 5

Exercise 1. In the class, we talked about different types of *pseudo-inverses*. In particular, we defined a *generalized inverse* or *g-inverse*: A^- is a g-inverse of A if $\mathrm{A}\mathrm{A}^-\mathrm{A} = \mathrm{A}$. We also saw that if A is $m \times n$, then for the definition to make sense, A^- must be $n \times m$. There is an alternate definition of a g-inverse:

> For a given matrix A, A^- is called its generalize inverse/g-inverse if the system of linear equations $\mathrm{A}\boldsymbol{\theta} = \mathbf{b}$ admits a solution $\widehat{\boldsymbol{\theta}} = \mathrm{A}^-\mathbf{b}$ for any $\mathbf{b} \in \mathcal{C}(\mathrm{A})$.

In the class, we saw that the former definition implies the latter definition (i.e., if $\mathrm{A}\mathrm{A}^-\mathrm{A} = \mathrm{A}$, then $\widehat{\boldsymbol{\theta}} = \mathrm{A}^-\mathbf{b}$ is a solution to $\mathrm{A}\boldsymbol{\theta} = \mathbf{b}$ for $\mathbf{b} \in \mathcal{C}(\mathrm{A})$).

(a) Verify that the latter also implies the former, i.e., if $\widehat{\boldsymbol{\theta}} = \mathrm{A}^-\mathbf{b}$ is a solution to $\mathrm{A}\boldsymbol{\theta} = \mathbf{b}$ for every $\mathbf{b} \in \mathcal{C}(\mathrm{A})$, then $\mathrm{A}\mathrm{A}^-\mathrm{A} = \mathrm{A}$.

Hint: $\widehat{\boldsymbol{\theta}} = \mathrm{A}^-\mathbf{b}$ *is a solution to* $\mathrm{A}\boldsymbol{\theta} = \mathbf{b}$ *means* $\mathrm{A}\mathrm{A}^-\mathbf{b} = \mathbf{b}$. *What happens if you take* $\mathbf{b} = \mathbf{a}^{(j)}$, *the* j*-th column of* A*?*

We also defined the Moore-Penrose g-inverse to be the matrix $\mathrm{A}^\dagger$ satisfying the *Penrose* conditions:

- $\mathrm{A}\mathrm{A}^\dagger\mathrm{A} = \mathrm{A}$.
- $\mathrm{A}^\dagger\mathrm{A}\mathrm{A}^\dagger = \mathrm{A}^\dagger$.
- $\mathrm{A}\mathrm{A}^\dagger$ and $\mathrm{A}^\dagger\mathrm{A}$ are symmetric.

A matrix satisfying only the first two conditions, i.e., $\mathrm{A}\mathrm{A}^-\mathrm{A} = \mathrm{A}$ and $\mathrm{A}^-\mathrm{A}\mathrm{A}^- = \mathrm{A}^-$, is called a *reflexive* g-inverse.

(b) Show that the Moore-Penrose g-inverse of a matrix is unique.

Hint: *Suppose two matrices* $\mathrm{A}_1^\dagger$ *and* $\mathrm{A}_2^\dagger$ *satisfy the Penrose conditions. Show that* $\mathrm{A}_1^\dagger\mathrm{A}\mathrm{A}_2^\dagger = \mathrm{A}_1^\dagger$.

(c) Suppose that $\mathrm{A}^\dagger$ is the Moore-Penrose g-inverse of A. Show that $(\mathrm{A}^\dagger)^\top$ is the Moore-Penrose g-inverse of $\mathrm{A}^\top$. Hence, deduce that if A is symmetric, then so is $\mathrm{A}^\dagger$.

In the following, we study some properties of g-inverses. Here, A^- denotes a generic g-inverse, which may be different from the Moore-Penrose g-inverse. We start by defining two other inverses.

Recall that the reason behind defining pseudo inverses was to solve linear equations: $\mathrm{A}\boldsymbol{\theta} = \mathbf{b}$. We may use $\widehat{\boldsymbol{\theta}} = \mathrm{A}^{\mathrm{L}}\mathbf{b}$, where A^{L} is a matrix satisfying $\mathrm{A}^{\mathrm{L}}\mathrm{A} = \mathrm{I}$. Any such matrix A^{L} is called a *left inverse* of A. Verify that for the definition to make sense, A^{L} must be of size $d \times n$ if A is $n \times d$.

(d) Show that A has a left inverse only if it is of full column rank.

Hint: $\mathrm{A}^{\mathrm{L}}\mathrm{A} = \mathrm{I}$ *implies* $\mathbf{e}_j \in \mathcal{R}(\mathrm{A})$ *for* $j = 1, \ldots, d$, *where* $\mathbf{e}_j = (0, \ldots, 0, \underbrace{1}_{j}, 0, \ldots, 0)^\top$.

(e) Check that if A is of full column rank, then $(\mathrm{A}^\top \mathrm{A})^{-1}\mathrm{A}^\top$ is a left inverse of A. Thus, A has a left inverse if and only if A is of full column rank.

(f) Verify that if $\mathrm{A}\boldsymbol{\theta} = \mathbf{b}$ is solvable, then $\widehat{\boldsymbol{\theta}} = \mathrm{A}^{\mathrm{L}}\mathbf{b}$ is a solution.

Another notion of inverse is that of a *right inverse*: A^{R} is a right inverse of A if $\mathrm{A}\mathrm{A}^{\mathrm{R}} = \mathrm{I}$. For a system of linear equations $\mathrm{A}\boldsymbol{\theta} = \mathbf{b}$, $\widehat{\boldsymbol{\theta}} = \mathrm{A}^{\mathrm{R}}\mathbf{b}$ is always a solution.

(g) Show that A has a right inverse only if A is of full row rank.

(h) For a full row rank matrix A, $\mathrm{A}^\top(\mathrm{A}\mathrm{A}^\top)^{-1}$ is a right inverse. Thus, A has a right inverse if and only if it is of full row rank.

Now, we are ready to study some properties of generalized inverses. We start by proving its existence. Recall that any $n \times d$ matrix A with $\mathrm{rank}(\mathrm{A}) = r$ can be written as $\mathrm{A} = \mathrm{BC}$, where B is an $n \times r$ matrix and C is an $r \times d$ matrix satisfying $\mathrm{rank}(\mathrm{B}) = \mathrm{rank}(\mathrm{C}) = r$. This is also known as the *rank factorization* of A. Notice that B has a left inverse and C has a right inverse.

(i) Show that $\mathrm{A}^- = \mathrm{C}^{\mathrm{R}}\mathrm{B}^{\mathrm{L}}$ is a g-inverse of A. Thus, every matrix A has a g-inverse.

(j) Now, consider $\mathrm{B}^{\mathrm{L}} = (\mathrm{B}^\top \mathrm{B})^{-1}\mathrm{B}^\top$ and $\mathrm{C}^{\mathrm{R}} = \mathrm{C}^\top(\mathrm{C}\mathrm{C}^\top)^{-1}$. Show that $\mathrm{A}^\dagger = \mathrm{C}^{\mathrm{R}}\mathrm{B}^{\mathrm{L}}$ is the Moore-Penrose g-inverse of A. Thus, every matrix A has a Moore-Penrose g-inverse.

(k) Show that $\mathrm{rank}(\mathrm{A}^-) \geq \mathrm{rank}(\mathrm{A})$. $\mathrm{rank}(\mathrm{A}^-) = \mathrm{rank}(\mathrm{A})$ if A^- is reflexive.

(l) Show that $\mathrm{rank}(\mathrm{A}\mathrm{A}^-) = \mathrm{rank}(\mathrm{A}^-\mathrm{A}) = \mathrm{rank}(\mathrm{A})$.

(m) Show that $\mathrm{A}\mathrm{A}^-$ and $\mathrm{A}^-\mathrm{A}$ are idempotent.

(n) Verify that $\mathrm{A}(\mathrm{A}^\top \mathrm{A})^-\mathrm{A}^\top \mathrm{A} = \mathrm{A}$ and $\mathrm{A}^\top \mathrm{A}(\mathrm{A}^\top \mathrm{A})^-\mathrm{A}^\top = \mathrm{A}^\top$.

   Hint: $\mathrm{B} = \mathrm{O}$ *if and only if* $\mathrm{B}^\top \mathrm{B} = \mathrm{O}$. *You may also use* $\mathcal{C}(\mathrm{A}) = \mathcal{C}(\mathrm{A}\mathrm{A}^\top)$ *and* $\mathcal{R}(\mathrm{A}) = \mathcal{R}(\mathrm{A}^\top \mathrm{A})$. *In particular, we can write* $\mathrm{A} = \mathrm{A}\mathrm{A}^\top \mathrm{W}$ *for some* W, *and* $\mathrm{A} = \mathrm{V}\mathrm{A}^\top \mathrm{A}$ *for some* V.

(o) Show that for matrices B and C satisfying $\mathcal{R}(\mathrm{B}) \subseteq \mathcal{R}(\mathrm{A})$ and $\mathcal{C}(\mathrm{C}) \subseteq \mathcal{C}(\mathrm{A})$, $\mathrm{B}\mathrm{A}^-\mathrm{C}$ does not depend on the choice of the g-inverse A^-.

   Hint: $\mathcal{R}(\mathrm{B}) \subseteq \mathcal{R}(\mathrm{A})$ *means the rows of* B *are elements of* $\mathcal{R}(\mathrm{A})$. *What does it say about the relationship between* B *and* A*? Similarly, try to find the relationship between* C *and* A.

(p) Use the previous part to show that $\mathrm{A}(\mathrm{A}^\top \mathrm{A})^-\mathrm{A}^\top$ does not depend on the choice of the g-inverse.

(q) Since $\mathrm{A}^\top \mathrm{A}$ is symmetric, there exists a symmetric g-inverse of it (e.g., the Moore-Penrose g-inverse). Therefore, in the previous part we can take $(\mathrm{A}^\top \mathrm{A})^-$ to be symmetric (as it does not depend on the choice). Use this to show that $\mathrm{P}_{\mathrm{A}} = \mathrm{A}(\mathrm{A}^\top \mathrm{A})^-\mathrm{A}^\top$ is an orthogonal projection matrix.

Next, we see a few different ways of finding g-inverses.

(q) Let $\mathrm{A} = \begin{pmatrix} \mathrm{A}_{11} & \mathrm{A}_{12} \\ \mathrm{A}_{21} & \mathrm{A}_{22} \end{pmatrix}$ be of rank r, where A_{11} is an $r \times r$ full-rank matrix. Then, $\mathrm{A}^- = \begin{pmatrix} \mathrm{A}_{11}^{-1} & \mathrm{O} \\ \mathrm{O} & \mathrm{O} \end{pmatrix}$ is a g-inverse of A. We did a partial proof of this result. A crucial step in that proof was to show $\mathrm{A}_{21}\mathrm{A}_{11}^{-1}\mathrm{A}_{12} = \mathrm{A}_{22}$. Consider the matrices $\begin{pmatrix} \mathrm{I} & \mathrm{O} \\ -\mathrm{A}_{21}\mathrm{A}_{11}^{-1} & \mathrm{I} \end{pmatrix}$ and $\begin{pmatrix} \mathrm{I} & -\mathrm{A}_{11}^{-1}\mathrm{A}_{12} \\ \mathrm{O} & \mathrm{I} \end{pmatrix}$. Notice that both of these matrices have determinant 1, so these are non-singular. Pre- and post-multiply A by these two matrices to see that the matrix $\begin{pmatrix} \mathrm{A}_{11} & \mathrm{O} \\ \mathrm{O} & \mathrm{A}_{22} - \mathrm{A}_{21}\mathrm{A}_{11}^{-1}\mathrm{A}_{12} \end{pmatrix}$ has rank r. Since A_{11} is of rank r, deduce that $\mathrm{A}_{22} - \mathrm{A}_{21}\mathrm{A}_{11}^{-1}\mathrm{A}_{12}$ must be O.

(r) Similarly, if $A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$ is of rank r with A_{22} an $r \times r$ full-rank matrix, then $A^- = \begin{pmatrix} O & O \\ O & A_{22}^{-1} \end{pmatrix}$ is a g-inverse of A.

Note: For any matrix A with rank r, we can always find an $r \times r$ submatrix which is non-singular. But, such a submatrix does not necessarily need to be in the A_{11} or A_{22} position. In such cases, we can extend the above construction as follows.

- Find any non-singular $r \times r$ submatrix C of A. The elements of C does not need to be adjacent.
- Find C^{-1} and its transpose.
- In A, replace the elements of C by the elements of $(C^{-1})^\top$. Replace all the other elements by 0.
- Transpose the resulting matrix.

(s) For a p.s.d. matrix A, we can write $A = U\Lambda U^\top$, where U is orthogonal and Λ is diagonal. If $\text{rank}(A) = r$, then $\Lambda = \begin{pmatrix} \Lambda_r & O \\ O & O \end{pmatrix}$. Then, for any P, Q, R, $A^- = U \begin{pmatrix} \Lambda_r^{-1} & P \\ Q & R \end{pmatrix} U^\top$ is a generalized inverse of A. The Moore-Pensore g-inverse is given by $A^\dagger = U \begin{pmatrix} \Lambda_r^{-1} & O \\ O & O \end{pmatrix} U^\top$.

(t) A similar result can be shown with the singular value decomposition. Any matrix A of rank r can be written as $A = U \begin{pmatrix} \Lambda_r & O \\ O & O \end{pmatrix} V^\top$, where Λ_r is an $r \times r$ diagonal matrix of rank r. Then, for any P, Q, R, $A^- = V \begin{pmatrix} \Lambda_r^{-1} & P \\ Q & R \end{pmatrix} U^\top$ is a g-inverse of A. The Moore-Penrose g-inverse is given by $A^\dagger = V \begin{pmatrix} \Lambda_r^{-1} & O \\ O & O \end{pmatrix} U^\top$.

(u) Next, we see a result similar to the inverse of a partitioned matrix, but now in terms of g-inverses. Consider a symmetric matrix A partitioned as $A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$. Show that the following is a g-inverse of A

$$A^- = \begin{pmatrix} A_{11}^- + A_{11}^- A_{12} B^- A_{21} A_{11}^- & -A_{11}^- A_{12} B^- \\ -B^- A_{21} A_{11}^- & B^- \end{pmatrix} = \begin{pmatrix} A_{11}^- & O \\ O & O \end{pmatrix} + \begin{pmatrix} A_{11}^- A_{12} \\ -I \end{pmatrix} B^- \begin{pmatrix} A_{21} A_{11}^- & -I \end{pmatrix},$$

where $B = A_{22} - A_{21} A_{11}^- A_{12}$.

Exercise 2 (Graded)**.** In the class, we gave a classification of different problems under the "linear models" framework, depending on the nature of the covariates $x_1, \ldots, x_d$ and the parameters $\theta_1, \ldots, \theta_d$:

$$x_i\text{'s} \begin{cases} \text{numerical} \rightarrow \text{regression} \\ \text{categorical} \rightarrow \text{ANOVA} \\ \text{mixed} \rightarrow \text{ANOCOVA} \end{cases} \qquad \theta\text{'s} \begin{cases} \text{fixed} \rightarrow \text{fixed effects model} \\ \text{random} \rightarrow \text{random effects model} \\ \text{mixed} \rightarrow \text{mixed effects model} \end{cases}$$

In this exercise, we will see a few examples with categorical covariates and how to handle them.

Suppose that we have a categorical covariate c with two categories c_1 and c_2. We can transform c into two numerical covariates x_1 and x_2 as follows

$$x_1 = \begin{cases} 1 & \text{if } c = c_1, \\ 0 & \text{otherwise,} \end{cases} \qquad \text{and} \qquad x_2 = \begin{cases} 1 & \text{if } c = c_2, \\ 0 & \text{otherwise.} \end{cases}$$

(a) Verify that the transformation $c \mapsto (x_1, x_2)$ is a bijection. That is, it is always possible to go back-and-forth between c and (x_1, x_2). So, there is no loss of information in making this transformation.

(b) Consider a linear model: $y = \alpha + \beta_1 x_1 + \beta_2 x_2 + \epsilon$. What do the parameters α, β_1 and β_2 signify? Check that the data matrix corresponding to this linear model is rank deficit. Can we estimate all the parameters of this linear model?

(c) Now consider a different transformation

$$x = \begin{cases} 1 & \text{if } c = c_1, \\ 0 & \text{if } c = c_2. \end{cases}$$

Show that the transformation $c \mapsto x$ is also a bijection. Consider the linear model $y = \gamma + \delta x + \varepsilon$. What do the parameters γ and δ of this model signify? What is the rank of the data matrix corresponding to this linear model? Is there any connection between the parameters (γ, δ) and the parameters $(\alpha, \beta_1, \beta_2)$?

(d) Suppose that instead of the transformation defined above, we use a different transformation

$$x' = \begin{cases} 0 & \text{if } c = c_1 \\ 1 & \text{if } c = c_2 \end{cases}.$$

This is also a valid transformation (after all, we have no preference over the categories!). What do the parameters of the model: $y = \gamma' + \delta' x' + \varepsilon'$ signify? Is there any connection between (γ, δ) and (γ', δ')?

(e) What would your conclusions in part (c) or (d) be for the following transformation?

$$x'' = \begin{cases} 1 & \text{if } c = c_1 \\ -1 & \text{if } c = c_2 \end{cases}.$$

(f) Formulate the following in terms of hypotheses tests involving the parameters of the models above.

i. The category c does not affect the response y.

ii. The effect of category c_1 is the same as that of category c_2.

(g) Can we extend the idea in part (a) if c has more than two categories? What about extending the idea in part (c) for more than two categories?

Note: In the above, $x_1 = x = \mathbf{1}_{\{c=c_1\}}$ and $x_2 = x' = \mathbf{1}_{\{c=c_2\}}$. Also, $x'' = x - x' = \mathbf{1}_{\{c=c_1\}} - \mathbf{1}_{\{c=c_2\}}$.

Exercise 3. In this exercise, we study a linear model with *interactions*. Suppose that we have two categorical covariates, denoting the presence or absence of two attributes A and B. For example, one can be whether a person is a smoker or not, while the other is the type of medicine given to the person. We can convert the categorical variables into numerical covariates as $c = \mathbf{1}_{\{A \text{ is present}\}}$ and $d = \mathbf{1}_{\{B \text{ is present}\}}$. Consider the following linear model with interaction:

$$y = \alpha + \beta_1 c + \beta_2 d + \beta_{12} cd + \epsilon, \qquad \epsilon \sim (0, \sigma^2).$$

The term β_{12} is called the interaction term.

(a) What do the parameters α, β_1, β_2 and β_{12} signify? Can you see why β_{12} is the interaction term?

Hint: *Write* $\mathbb{E}(y)$ *in the four cases: A absent/present, B absent/present.*

(b) Formulate the following into statistical hypotheses in terms of the model parameters.

   i. Presence or absence of A does not affect the response y.
   ii. Presence or absence of B does not affect the response y.
   iii. The effect of A are B are the same.
   iv. The effect of B is thrice the effect of A.
   v. The effect of A is unchanged by the presence or absence of B.

(c) Repeat the exercise with one numerical and one categorical covariate with two categories. That is, $c = \mathbf{1}_{\{A\}}$ as before and x is a numerical variable. For example, A can be a medication and x can be the age of the person. What do the parameters of the model $y = \delta + \gamma_1 c + \gamma_2 x + \gamma_{12} cx + \varepsilon$ signify? Formulate the following into statistical hypotheses in terms of the model parameters.

   i. Presence or absence of the attribute A does not affect the response.
   ii. The covariate x does not affect the response.
   iii. The effect of the covariate x on y is unaffected by the presence or absence of A.
   iv. The effect of x in the presence of A is five times that of the effect of x in the absence of A.

Exercise 4 (Simulation Study 5)**.**

In `R`, the `lm` function can be used to fit a linear model with categorical covariates. In this exercise, we will study what `lm` does when it encounters a categorical covariate.

(a) Simulate a dataset of size n with response y and a single categorical covariate x with two categories c_1 and c_2. For instance, you can use `x <- c(rep("cat", 20), rep("dog", 20))` to get a categorical covariate x with two categories `"cat"` and `"dog"`. Now, you can assign some real numbers as your response, e.g., `y <- rnorm(40)`.

(b) Fit a linear model using `model <- lm(y ~ x)`.

(c) Use `summary(model)` to get the model summary. What do you see? Try to write down the model fitted by `R` in terms of x and y. Is it possible to estimate the effects of both "cat" and "dog" from this summary?

(d) Now, let us fit a linear model without intercept to this data. A model without the intercept can be fitted using `lm(y ~ 0 + x)` or `lm(y ~ x - 1)`. Check the summary. Write down the fitted model. Do you see any connection to the previously fitted model? Should there exist any connection between the two models?

(e) Repeat the exercise with a categorical covariate with three or more categories. What do you observe?

(f) Now repeat the exercise with two categorical covariates. You may also add an interaction term in your model. An interaction term between two covariates x_1 and x_2 can be added using `x1:x2` in the linear model formula. Try to write down the model being fitted. Identify the parameter estimates and connect them to your model.

Exercise 5 (Simulation study 6)**.** In the class, we have seen that we cannot measure the effect of the covariate in a simple linear regression model $y_i = \alpha + \beta x_i + \epsilon_i$ if all the x_i values are the same. For instance, it is not possible to estimate the effect of `age` on `blood pressure` in the model `pressure` $= \alpha + \beta \times$ `age` $+ \epsilon$ using data from 21-year old people only. Let us see what would `R` do for such a dataset.

(a) Simulate a dataset of n (of your choice, e.g., 100) individuals, all of whom are 21 years old. In `R`, you may use `rep(a,n)` to get a vector of size n with all the elements equal to a. You can use the strategy in the previous exercise to generate observations for `blood pressure`. Use the `lm` function in `R` to fit a linear model with intercept to your generated data. What do you see? Use `summary` to check the results.

(b) We have also seen that we won't have any issues if there are at least two individuals with different ages in the data set. Add one more observation with `age` $= 22$ to your dataset and repeat part (a) with this new data set. What do you see in this case?

(c) Now, suppose that you were able to determine how to collect the data. It seems reasonable to have some variability in the `age`-s of the individuals in order to measure the effect of `age` on the response. Let us see how this idea is manifested in the results.

  i. Generate n `age`-s in $[18, 24]$ and repeat the simulation study as above. In `R`, you can use `runif` to generate `uniform` random variable within a given range. Compare these results with the results from the previous simulations.

  ii. Now consider different age brackets $[a, b]$ and repeat the simulation study. What is the effect of the age bracket on the result? Do you see any difference in the results for larger or smaller age brackets?

  Note: Keep the β parameter fixed in all the simulation setups while simulating the response.

  Note: For an estimator, it is good if the estimated parameter value is close to the actual value (since we know it in simulations). But also keep an eye on the standard error of the estimator, which signifies how precise the estimate is.

  iii. Based on your observations above, how would you design an experiment for the given problem? That is, how would you decide which individuals to select for the given study?